

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

What drives a happy customer vs. a one-star review; and where the marketplace should invest to grow without breaking the experience.

Olist Brazilian E-Commerce dataset + Marketing Funnel dataset · 99,441 orders · Sept 2016 – Aug 2018

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Delivery Performance

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Product & Pricing

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Seller Performance

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Regional Logistics

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Seller Behaviour & Investment

THE PROBLEM

Growth and satisfaction pull in opposite directions

Onboard sellers aggressively and gross sales rise; but late deliveries, poorly-rated products, and bad reviews start driving customers away. Crack down on every slow seller and quality improves; but the catalogue shrinks and growth stalls. Every decision leadership makes trades growth against satisfaction, and right now those calls are made on gut feel.

14.6%

of reviewed orders end in a 1–2 star review

96,478

orders delivered, of 99,441 placed

3,095

sellers on the platform

THE GUIDING QUESTION

Which product categories, seller behaviours, regions, and delivery patterns predict a happy customer vs. a one-star review; and where should the company invest to grow the marketplace without breaking the customer experience?

The average review hides one unhappy customer in seven



1–2 stars together: 14.6% of 98,673 reviewed orders, one in seven. The 4.09 average hides them, so every analysis here tracks the negative rate.

9 tables → 1 row per order

Items, payments and reviews are aggregated to order level before joining, so a multi-item order is counted once, not once per item. 99,441 orders.

One review per order

202 orders had two different scores. We tried all four ways of picking one: the negative rate moves by less than 0.1 pp, so the choice doesn't matter.

Failures kept, not deleted

2,849 reviewed orders have no delivery date; nearly all never arrived. Reported separately, because they carry 15% of all bad reviews.

Five members, five viewpoints, one shared definition of a bad experience

A “bad experience” is a review of 1–2 stars out of 5. Every analysis in this deck reports the negative rate (the share of 1–2 star reviews), never the average score, which hides them.

DELIVERY

Yash Arabhavi

Delay → reviews · Slow vs. broken promise

Anushka

Regional performance · Worst routes

SELLERS

Kannan

Damage concentration · Repeat offenders

Sahib Randhawa

Behavioural drivers · Where to invest

PRODUCT

Ashwanth

Beyond delivery · Category quadrant

Missing the promised date is a cliff, not a slope

9.2% → 62.3%

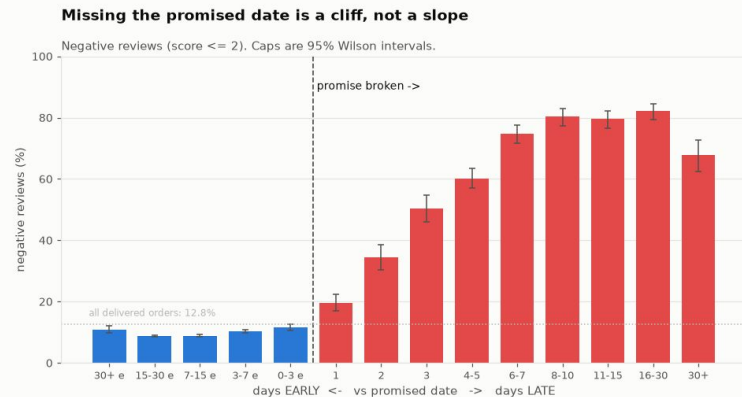
On-time orders are 9.2% negative; late orders are 62.3% negative; 6.8× worse.

6.7% of orders, 32.5% of bad reviews

Late deliveries are a small share of delivered orders but drive a third of their negative reviews.

Flat before, cliff after

However early it arrives: 8.9–11.6% negative. One day late: 19.6%. Three days late: 50.4%.



Olist Brazilian E-Commerce, 2016-2018

CAVEAT

2,849 reviewed orders have no delivery date; nearly all never arrived (78% negative). With no delivery date there is no delay to measure, so they sit outside this chart, not outside the story.

It's the broken promise, not the wait, that angers customers

21.2% vs 25.4%

A 31–45 day wait that arrived on time (21.2% negative) still beats being just 1–2 days late (25.4%). The gap is significant ($p = 0.02$).

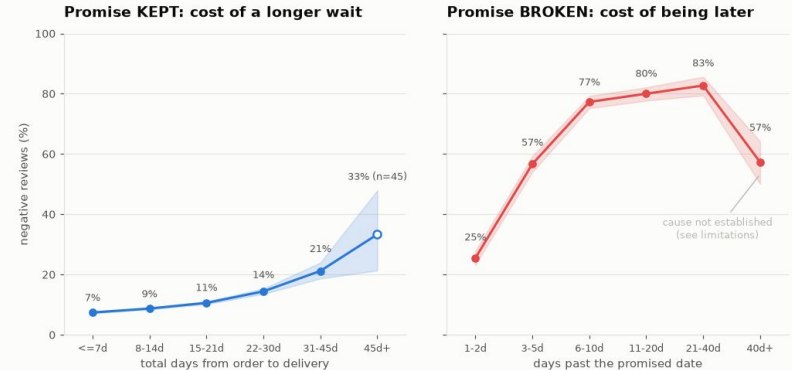
74.5% vs 0.3%

If nothing has arrived, the survey is sent on the promised date: 74.5% of late customers rated before their parcel came (on time: 0.3%).

RECOMMENDATION

Customers rate the gap between what they were told and what happened. Fix the promise, not the couriers: the estimate is the platform's own number, so it needs no logistics spend. Make it accurate, not just longer: the median order already arrives 12 days early, and the misses cluster on specific routes.

Being slow costs points. Breaking the promise costs the customer.
One shared scale. A month-long on-time wait still beats missing the date by two days.



Olist Brazilian E-Commerce, 2016-2018

Cost of waiting when the promise holds (left) vs. when it's broken (right); same shared scale

AL and MA are the clearest regional delivery-risk hotspots

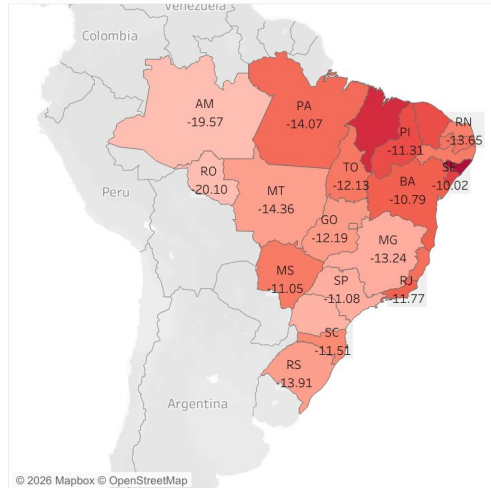
21.4% late in AL

Alagoas has the highest late-delivery rate among reliable states, at 21.4%, while its average review score is only 3.86.

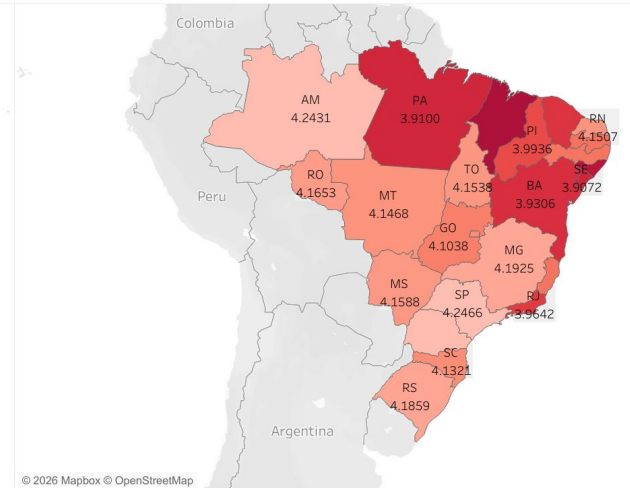
MA combines lateness + lowest satisfaction

Maranhão records 17.4% late deliveries and the lowest average review score (3.83) among the reliable states.

States by Late Delivery Rate



States by Average Review Score



RECOMMENDATION

Prioritize AL and MA for operational review. Investigate carrier performance, fulfillment lead times, distance, handoff delays and estimated-delivery accuracy for these destinations.

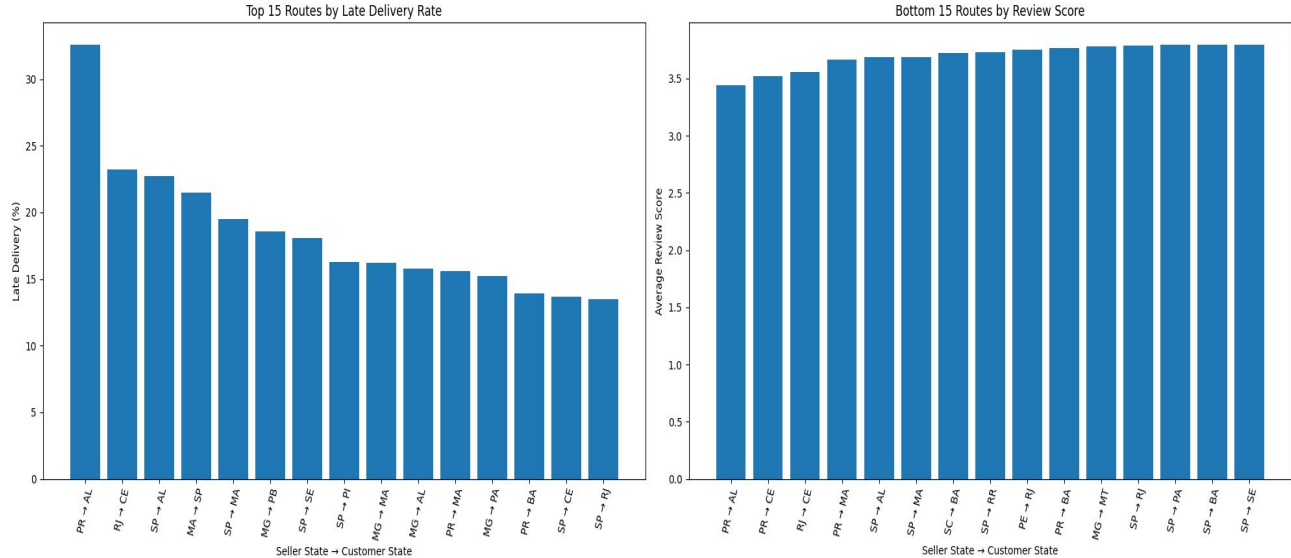
PR → AL is the worst route, while SP → AL and SP → MA create the biggest volume risk

32.6% late on PR → AL

PR → AL is the strongest route hotspot, with 32.6% late deliveries and the lowest route review score of 3.44.

High-volume routes amplify the problem

SP → AL has 22.7% late deliveries across 256 orders, while SP → MA has 19.5% across 493 orders — making them potentially more impactful than smaller routes.



RECOMMENDATION

Create route-level exception monitoring for PR → AL, RJ → CE, SP → AL and SP → MA, using both late-delivery percentage and review score.

The worst 5% of sellers cause 58% of bad reviews — but they also handle 53% of orders

+5pp, not +58pp

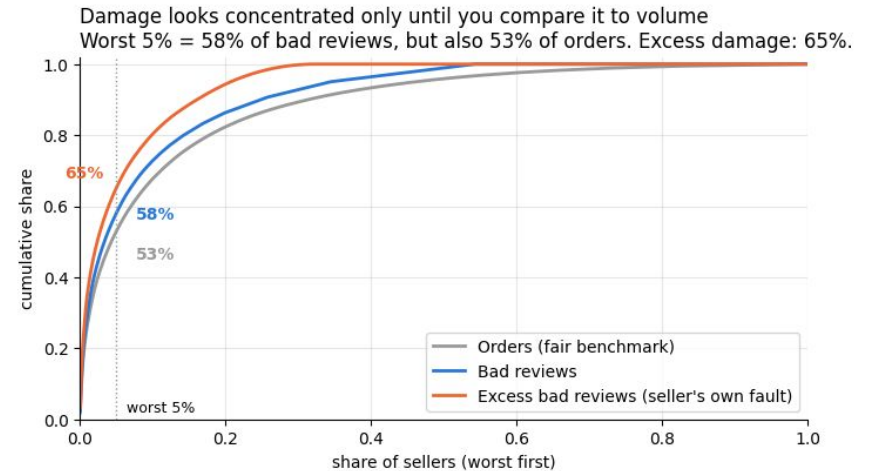
The worst 5% by bad-review count handle 53% of all orders — so their real excess over a fair share is only +5 percentage points.

65% of avoidable damage

Once category and delivery effects are removed, the worst 5% of sellers own 65% of the bad reviews their record can't explain.

WHY THIS MATTERS

The headline “worst 5% cause 58% of bad reviews” is misleading on its own — it mostly rediscovers seller size. Acting on raw counts would target the platform's biggest partners. Excess damage is the number to act on.



Cumulative share of orders vs. bad reviews vs. excess (avoidable) damage, sellers ranked worst-first

Yes — 77 sellers are repeatedly worse than their category and delivery record predicts

2.4× worse, even on time

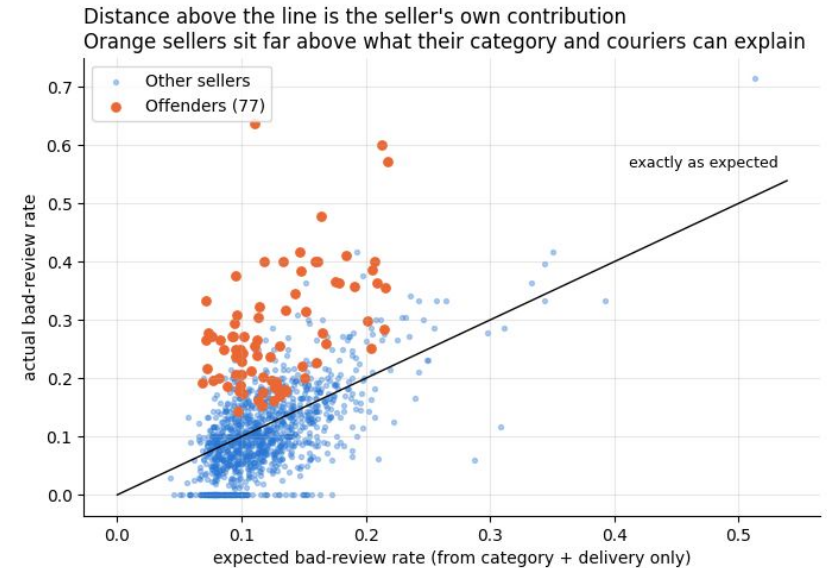
Removing the logistics excuse entirely, these 77 “offenders” draw an 18.4% bad-review rate vs. 7.6% for everyone else.

20× more than chance

33 sellers sit 3+ standard deviations above their own expectation — pure random noise would produce fewer than 2.

10% of GMV, 53% of avoidable damage

A small, identifiable group responsible for the majority of the damage that category and delivery can't explain.



Every seller's actual bad-review rate vs. what its category + delivery record predicts

Coach the 77, don't delist them

Coach to median: -1.30pp, 0% GMV lost

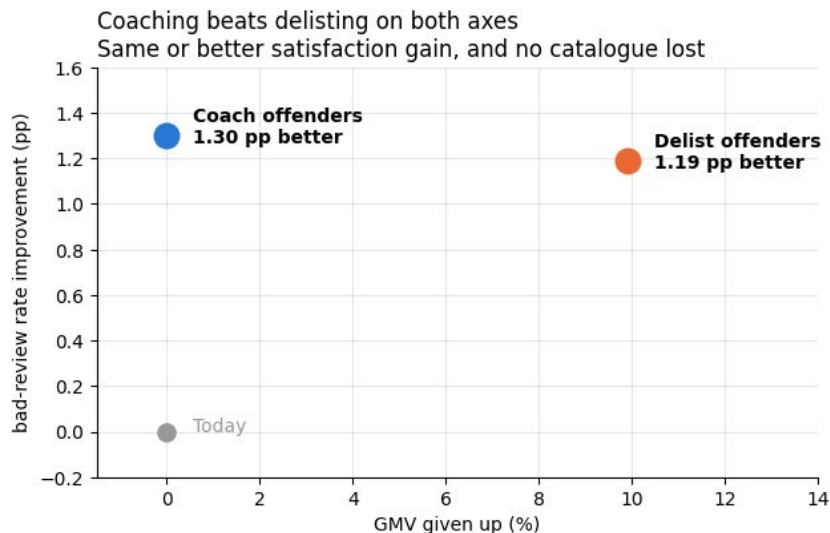
Bringing the 77 offenders to the median seller's performance improves the platform bad-rate with no revenue given up.

Delist instead: -1.19pp, -9.9% GMV

A smaller satisfaction gain, and it costs a tenth of platform revenue — delisting removes the ~78% of an offender's orders customers were happy with too.

SPLIT BY FAILURE MODE

26 of the 77 ship fast (3.7% late — better than average) but the item still disappoints. No shipping SLA fixes this group; they need a listing and quality review instead. Delisting should be a last resort, not the opening move.



Platform bad-review-rate improvement vs. GMV given up, for each intervention

Handling time is the dominant driver — volume and freight pricing are not reliable predictors

$r = +0.24, p < 0.0001$

Handling time is the only behaviour with a strong, highly significant correlation to a seller's excess bad-review rate.

Volume & freight: not significant

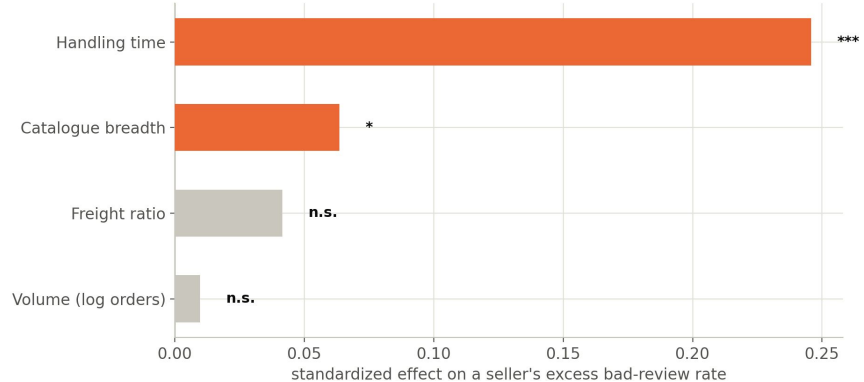
Order volume ($r=+0.02$) and freight ratio ($r=+0.04$) show no reliable link to excess bad reviews, controls held.

Robust to controls

The effect survives a multiple regression ($\beta=0.0079, p<0.0001$) and holds on-time orders only.

Handling time dominates the other three behaviours combined

Standardized regression coefficients (category + delivery band already netted out via the excess measure)



WHY THIS MATTERS

Catalogue breadth has a small, marginal effect ($p=0.035$) but is dwarfed by handling time — seller-quality investment should prioritize handling-time coaching over volume caps or freight rules.

Fund the zero-cost delivery-promise fix first, then sequence seller coaching by handling time

3.56pp vs 1.30pp

The delivery-promise fix outweighs seller coaching roughly 2.7 to 1 — both cost nothing in lost GMV.

77 sellers, coached not delisted

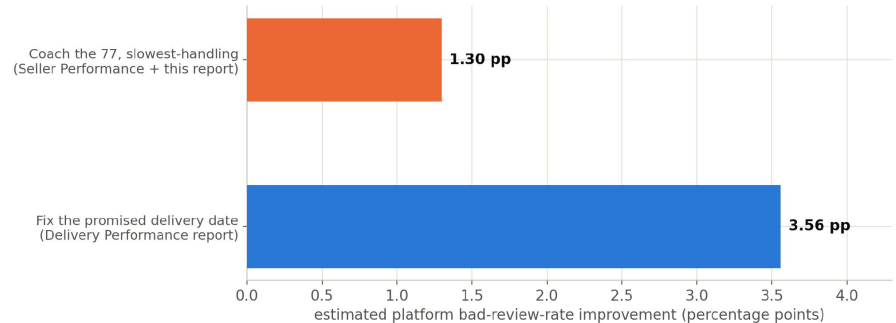
Coaching the 77 flagged sellers to the median cuts bad reviews with 0% GMV given up — sequence them by handling time.

Two levers not yet sized

Category/freight fixes (58% of bad reviews from 10 categories) and regional fixes are real but not yet sized in pp terms.

The two quantified levers, side by side

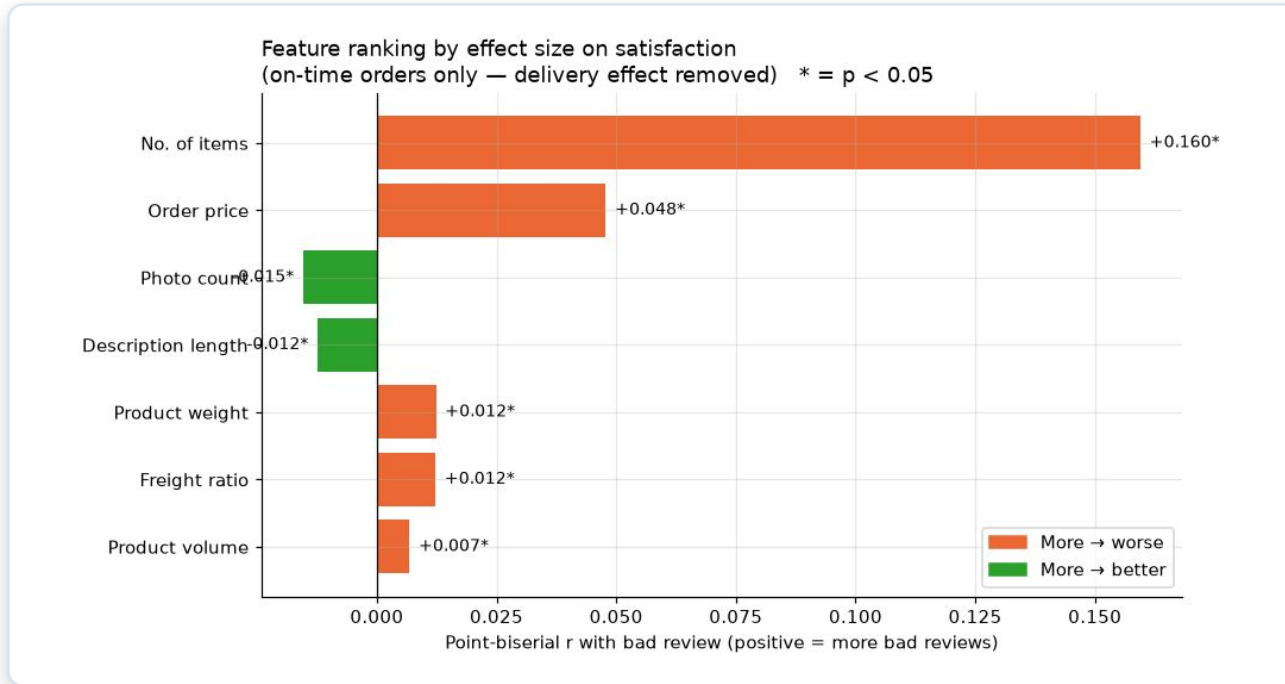
Category/freight fixes and regional fixes are real but not yet sized in pp terms



RECOMMENDATION

Fix delivery honesty first (~3.6pp, zero cost), then sequence seller coaching by handling time (1.30pp) — extend the other two findings to a matching pp estimate before ranking them.

Beyond delivery looking into dimensions of products and orders



Visual: Correlation between each product feature and is_bad (1-2 star review) as 0 or 1.

Computed on 89,443 on-time orders only

orange = more of it → worse;
green = more of it → better;

Outcome: order size (items) and price are the strongest secondary drivers of dissatisfaction

Q1 • Price and freight ratio

Bad-review rate across bins of order price and freight ratio, on-time orders only.

Order price (R\$, log bins)



Freight ratio



■ Order price ■ Freight ratio

High price orders gains more review

Maybe user engages more when the order price is more.
This may be applicable to positive reviews also.

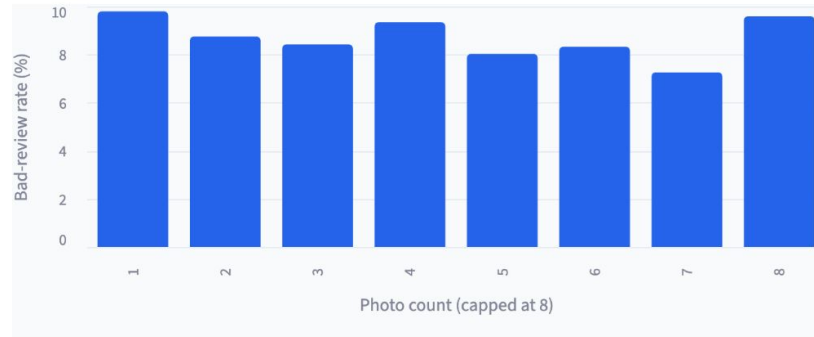
Freight ratio: strongest price signal

Orders where freight exceeds ~50% of item price see nearly 2× the bad-review rate — customers feel the shipping cost is unfair.

Q1 • Photo count and number of items

More product photos (up to ~4) are associated with fewer bad reviews; multi-item orders have more failure points.

Photo count (capped at 8)



■ Bad-review rate (%)

Number of items (capped at 5)



Photos & descriptions help

More photos (up to ~4) and longer descriptions modestly reduce bad reviews by setting accurate expectations.

No. of items: strongest driver

Multi-item orders have more points of failure — 4+ items: 32% negative vs. 8% for single-item orders.

Ten categories carry over half of revenue with below average satisfaction

High Rev + Poor Sat (fix urgently)

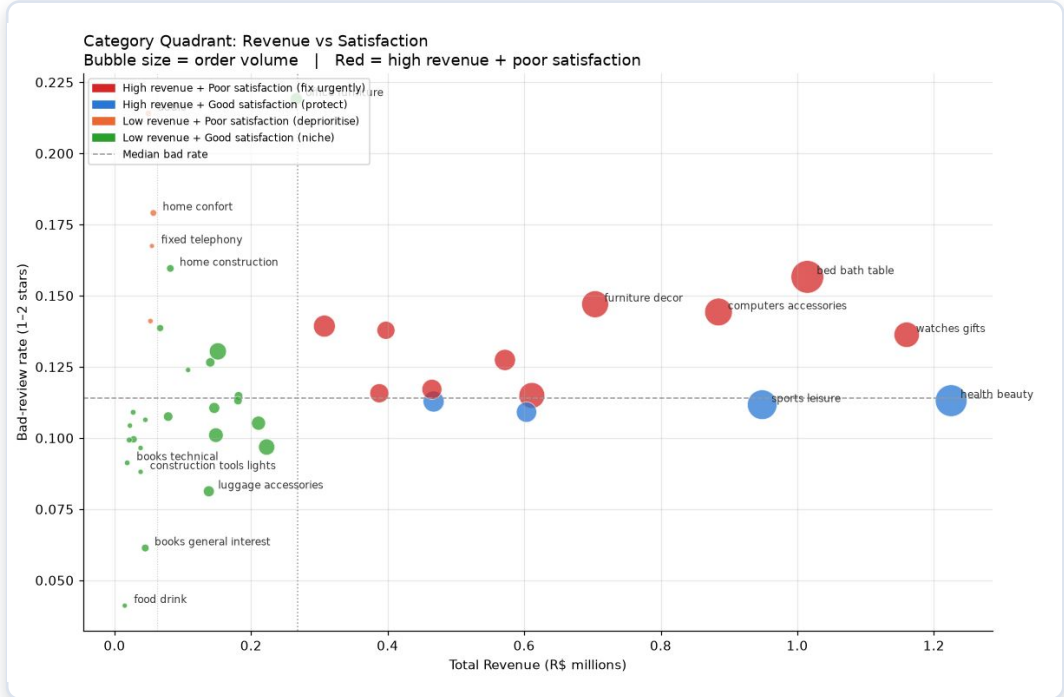
10 categories = 52.5% of revenue, 58.2% of bad reviews.
Watches & gifts, bed/bath/table, computers accessories, furniture décor...

High Rev + Good Sat (protect)

Health & beauty, sports & leisure, toys, cool stuff — the platform's engine. Monitor closely as they scale.

RECOMMENDATION

Apply the Q1 fixes (photo counts, freight transparency, description length) to the 10 red categories first that's where quality investment has the highest revenue-at-risk payoff.



Revenue vs. satisfaction by category — bubble size = order volume

LIMITATIONS & NEXT STEPS

Further Research Analysis

- **Patterns, not proof**

Every finding here is an association established through careful controls (delivery, category) — not a causal model.

- **Silent churn is invisible**

Orders that never arrive are 2.9% of reviewed orders but 15.3% of all bad reviews, and reviews only come from customers who respond to the survey at all.

Recommendations

- **Fix the promised delivery date**

It's a zero-cost change (it's just making the platform's own delivery estimate more honest/accurate), and it's the single biggest lever anyone on the team found — an estimated 3.6 percentage point drop in the platform's bad-review rate. Since it doesn't cost any revenue or require touching sellers at all, there's no reason not to fund it first.

- **Coach underperforming sellers**

Coach the 77 flagged sellers instead of delisting them, starting with the slowest-handling ones first. Coaching gets a 1.30pp improvement at zero revenue cost, versus delisting's 1.19pp while giving up ~10% of GMV — and since handling time is the strongest predictor of their bad reviews, that's where coaches get the fastest wins.

Thank you — Group 2